

Unit ID: 679b18f0aaf593b2823f3555

Unit 3 - Creating a Digital Ruler

PDF Documents

PDF Document 1: 1738217710396-Chapter 4 - Unit 3 - Creating a Digital Ruler.pdf

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Update #1 - 2025-05-16 15:27:43



UNIT 3 CREATING A DIGITAL RULER

A digital ruler is designed to measure more accurately and conveniently than a traditional ruler.

HOW DOES IT WORK

This hands-on project uses a ToF (Time-of-Flight) sensor to create a simple digital ruler. With the ToF sensor, you'll measure lengths and add an extra feature to switch between centimeters and inches as units.

The need for such digital measuring tools is becoming important for the following reasons:

1 ACCURACY AND CONSISTENCY

Digital rulers provide higher accuracy than analog rulers. Traditional rulers require aligning the markings with the human eye, which can lead to minor errors. In contrast, a digital ruler uses an electronic sensor to digitize measurements, allowing for precise readings down to tiny units. This high accuracy is essential for fields like machine part measurement or scientific experiments, where even a slight error can have significant effects.

2 QUICK MEASUREMENT AND EFFICIENCY

Digital rulers display measurements instantly, eliminating the need to read and align markings manually. With traditional rulers, checking measurements takes time, but digital rulers allow quick measurement, significantly improving work efficiency. In situations where multiple objects need to be measured quickly, digital rulers are much more efficient.

3 DATA STORAGE AND TRANSFER

Many digital rulers can store measurements or transfer data to other devices. For example, if you need to measure multiple objects and record the results continuously, a digital ruler can automatically save data, reducing the need for manual recording. This function saves time, especially in research or design work. Stored data can be transferred to a computer, making analysis and storage easier.

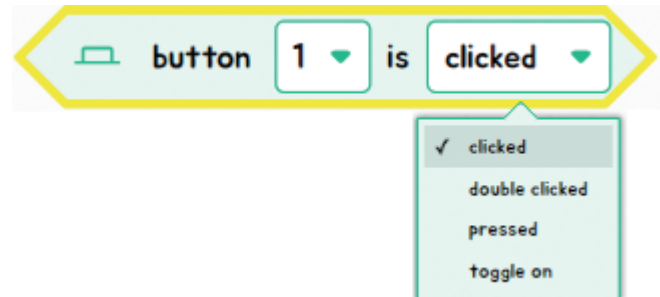
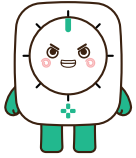
4 CONVENIENT UNIT CONVERSION

Digital rulers support multiple units, such as meters, centimeters, millimeters, and inches, allowing users to switch units with a single button press. Traditional rulers require manual conversion, but digital rulers make unit conversion easy, which is particularly useful in everyday life. This is especially helpful in international projects, where accurate unit conversion helps avoid confusion.



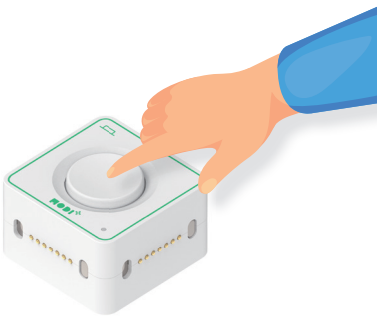


Do you remember which action Button can received?



Button can receive 4 types of information.

1. CLICK



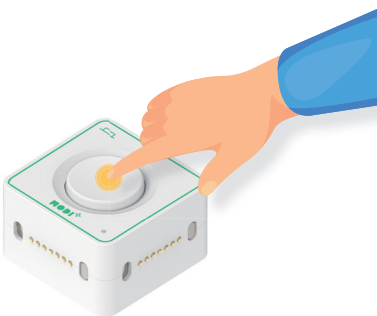
Detects if Button is clicked or not.

2. DOUBLE CLICK



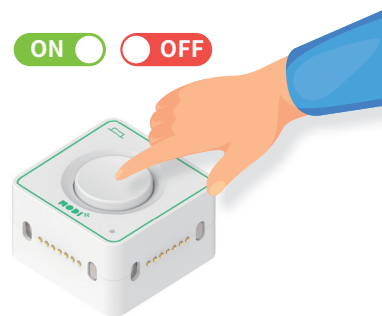
Detects if Button is double - clicked or not.

3. PRESS



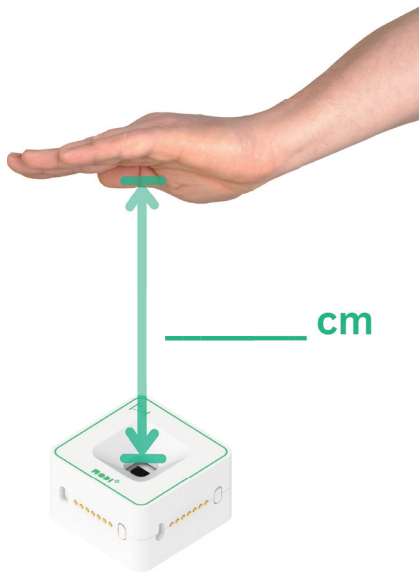
Show the press status of Button.

4. TOGGLE



Changes on / off when Button is clicked.

Key Features of ToF (Time-of-Flight) Sensor



DISTANCE MEASUREMENT

The ToF sensor measures how far away something is by sending a light signal that bounces off an object and returns to the sensor. The longer it takes, the farther the object is.

REAL-TIME PROCESSING

The sensor measures distance instantly, which is useful for moving environments, like robots or self-driving cars, where quick reactions are needed.

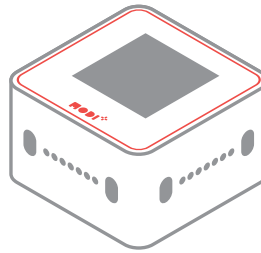
Material Needed



Network



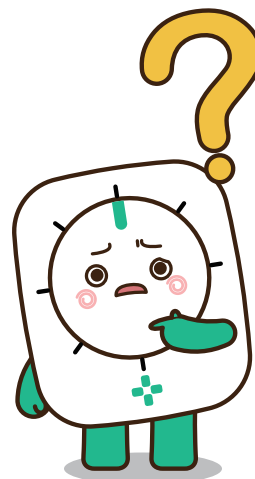
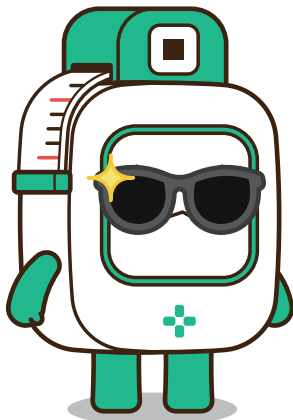
Dial



Display

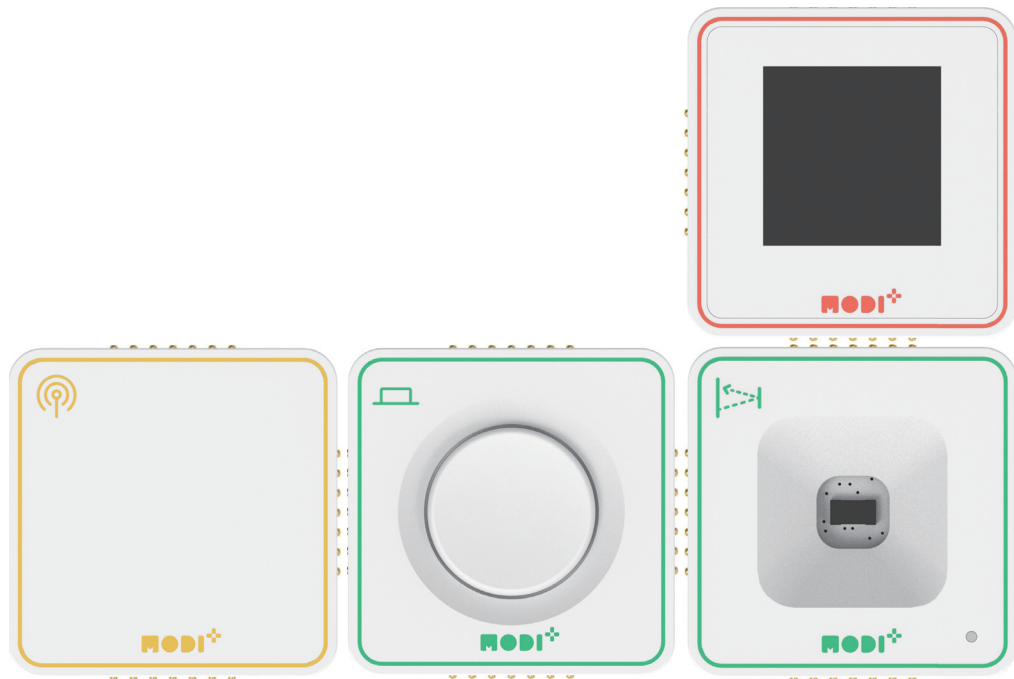


Button





When we combine all the modules, a creation has to look like this.



Code Story

If the toggle button is on

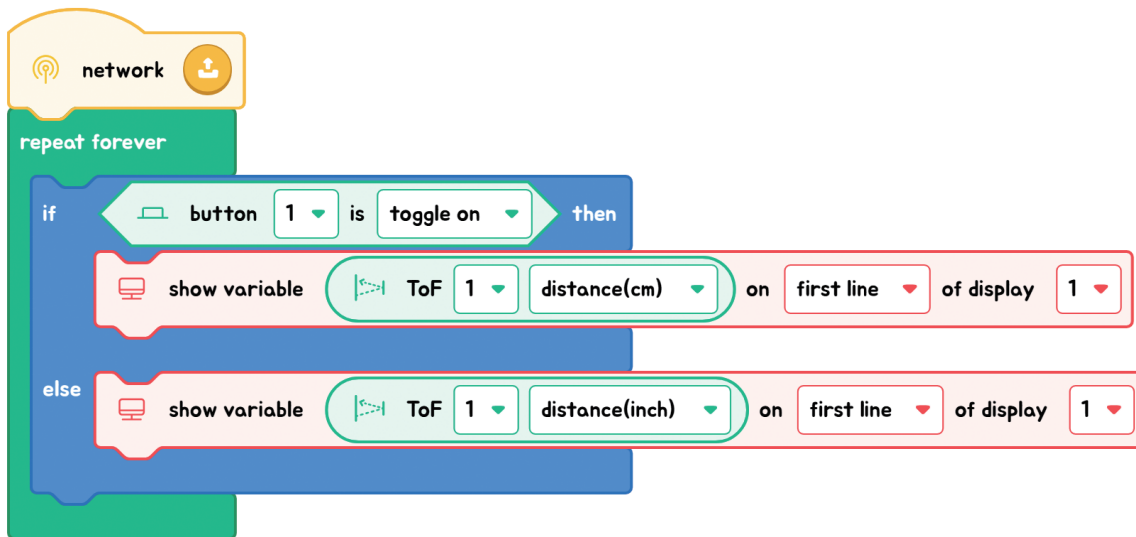
↓ true

The result in **"cm"** is displayed on the monitor.

↓ false

The result in **"inch"** is displayed on the monitor.

Code Quest



Checkpoint

- 1 Use the digital ruler to measure the width and height of Motor A in inches.
- 2 To prevent difficulty due to real-time changes, adjust the digital ruler so that the measurement is only taken when the button is clicked. Additionally, set up a double-click function to switch units between centimeters and inches.

